(a) Encrypts the input data and stores it as a file
(b) Decrypts the data it reads from a file and displays it
Can you write a small code snippet to do the same? If you are told that your program will execute on a processor, which supports AES-NI instructions, how will you modify your code to take advantage of this functionality?

12. Given an array of N integers and a number 'k', write a function which returns true if the array has a contiguous sub-sequence, which sums up to exactly 'k'? How would your code change if you are asked to return the start and end of the contiguous sub-sequence, if such a sub-sequence exists in the array. What is the time complexity of your solution?
13. You are given an array of N integers and asked to return a popular item in the array, if such a popular item exists in the array. A popular item is an element which repeats at least N/4 times in the array. How would you modify your code if you are asked to return all such popular elements? What is the time and space complexity of your solution?
14. You are given an array of N integers and asked to write a program to find the maximum difference between two array elements A[i] and A[j] such that i < j. How would your solution change if the condition that 'i' needs to be smaller than 'j' need not hold true for your program?
15. You are given an array of N elements, where each element is a task. Each task has a priority, which is marked as either *High*, *Medium* or *Low*. You are asked to rearrange the array such that all tasks which have the priority value 'High' appear first, followed by tasks whose priority is 'Medium', and then by tasks whose priority is 'Low'. You are asked to do this in-place with constant additional storage. What is the time complexity of your solution?
16. You are given a text file, which contains C/C++ code. The code has comments in it. The comments can be either single line or multi-line. Single line comments begin with a '//' character and do not span beyond a line. Multi-line comments are enclosed by '/*' and '*/' characters. Comments cannot nest inside each other. You are asked to write a code snippet which strips out the comments from the code in the input file.
17. You are given a binary search tree containing N integers. You are asked to return the 'kth' largest element in the binary search tree. Write a code snippet to do this. How would your solution change if you are asked to return the 'largest' element and the 'smallest' element in the binary search tree? What is the time and space complexity of your solution?
18. You are given a list of N users along with their login and logout times on the computer, over a period of one week. You are asked to write a program to:
    a. Find the time when the number of users that were logged into the computer was the maximum
    b. Find the time when the number of users that were logged into the computer was the minimum
    c. Find the user who was online for the maximum cumulative time over the one week
    d. Find the user who was online for the longest single duration slot over that week

    What is the data structure you would use to represent the login, logout time intervals?
19. A string is said to be unique if no two characters in the given string are identical. Given a set of N strings, you are asked to find all unique strings present in the set.
20. Given a string, find all the sub-strings of that string which are palindromes.
21. You are given two binary trees and asked to determine whether the two trees are identical both in terms of structure and the elements contained in them. Write a program to do this. What is the time and space complexity of your solution? How would your code change if you are told that the binary trees are actually binary search trees?
22. If you are asked to write a sort routine to replace the sort utility in UNIX/Linux, which sorting algorithm would you use and why?
23. If you are asked to design a data structure which needs to support (a) insert, (b) delete, (c) search, (d) find-minimum, and (e) find-maximum, which data structure would you use and why?
24. What is a lock-free data structure? Can you give an example of one?
25. The two terms 'concurrency' and 'parallelism' are often mistaken for each other in computer science. Can you differentiate between the two with appropriate examples of where concurrency exists and where parallelism does?

If you have any favourite programming questions or software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at *sandyasm_AT_yahoo_DOT_com*. Till we meet again next month, happy programming! END

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